Machine Learning

Statistical Learning: Bayesian Classification

Hoda Eldardiry

Lecture Objectives

- Explain learning and inference from a probabilistic perspective
- Distinguish joint, marginal, and conditional distributions
- Apply Bayes' rule to reverse conditionals
- Interpret Bayesian learning as updating beliefs
- Use a simple Bayesian model to make predictions

What does “learn from data” mean?

- ***ML = manage uncertainty using data***
- Real world is uncertain → probability models uncertainty
  - *Probability distribution quantifies uncertainty*
- Learning = reduce uncertainty
- Bayesian view: update beliefs when you see data
  - *Learning reduces uncertainty by updating probability distribution*

Probability Distributions Refresher

- **Random variables** represent quantities of interest
- **Probability distribution = uncertainty over outcomes**
    - Specifies probability for each event in a sample set
    - *Example: Uncertainty when rolling a dice over 6 possible outcomes*
        - Probabilities must sum to 1
- **Joint Distribution = world described by multiple variables**
    - Specifies probability of *combinations* of values
    - *Example: Weather: temperature, wind speed, humidity*
    - Probability that each takes certain values
- **Marginalization (Sum-out Rule)**
    - Why marginalize? We don't always care about everything
    - **Intuition:** collapsing dimensions

Joint → Marginal

- Given: joint distribution over random variables A: temp, B: humidity

$$P(A = a \wedge B = b) \forall_{a,b}$$

random variable ↗ value ↗

- Goal: Make a prediction for only one random variable (A: temp)

- Marginalization (sum-out rule) = extract distribution of one variable

$$P(A = a) = \sum_b P(A = a \wedge B = b)$$ Sum-out all possible values for B

$$P(B = b) = \sum_a P(A = a \wedge B = b)$$ Sum-out all possible values for A

Example

sunny

	cold	~cold
headache	0.108	0.012
~headache	0.016	0.064

~sunny

	cold	~cold
headache	0.072	0.008
~headache	0.144	0.576

$P(headache \wedge sunny \wedge cold) =$

$P(\sim headache \wedge sunny \wedge \sim cold) =$

$P(headache \vee sunny) =$

$P(headache) =$

Joint Distribution - Example

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$P(headache \vee sunny) =$ 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28

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marginalization

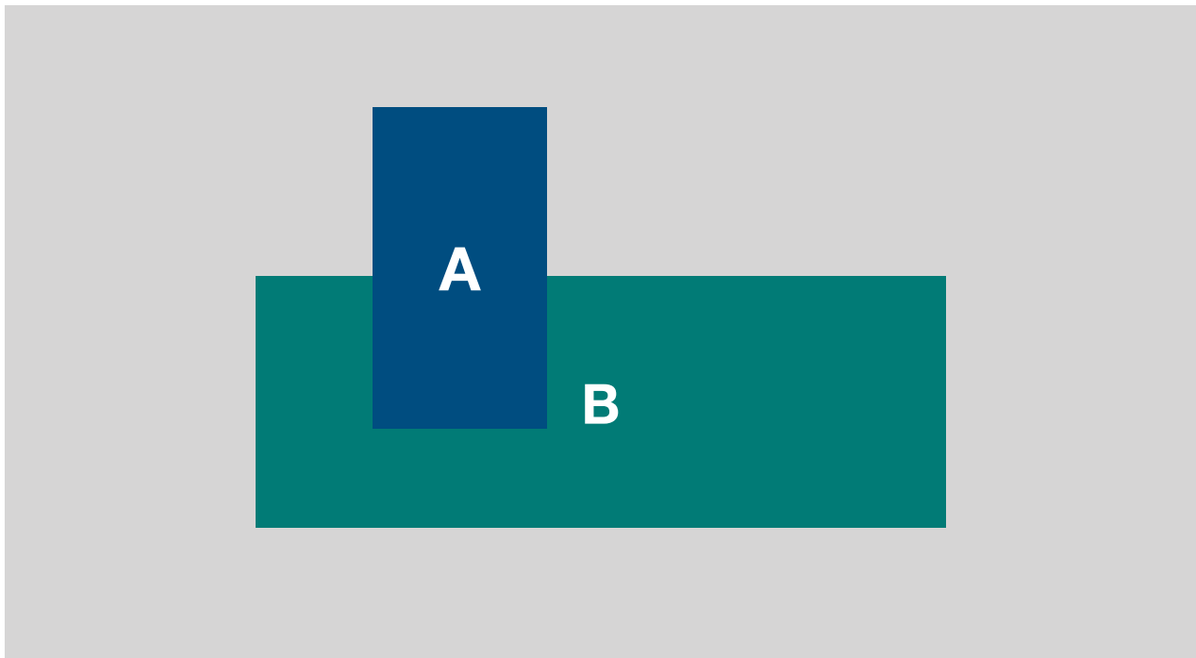
"Which variables are being summed out here, and why?"

Types of distributions

- Joint probability distribution
- Marginal probability distribution
- **Conditional probability distribution**

Conditional Probability

- **Definition:** P(A | B): probability of A *given* B is true
- **Interpretation:** restricted world
- **Formula:** P(A | B) = P(A ∧ B) / P(B)
- **Visual intuition:** Area diagrams (restricted sample space)
  - Let's see an example



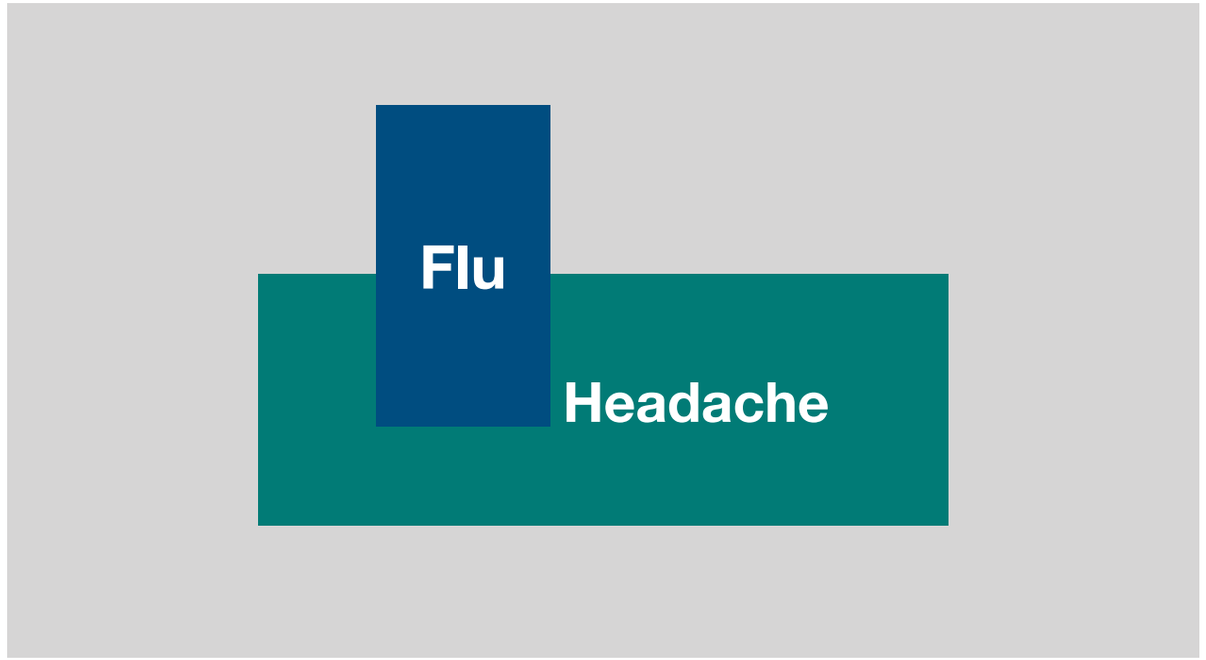
Flu-Headache Example

- *You wake up with a headache. Do you have the flu?*

- ***Priors:***

- P(H) = 1/10

- P(F) = 1/40

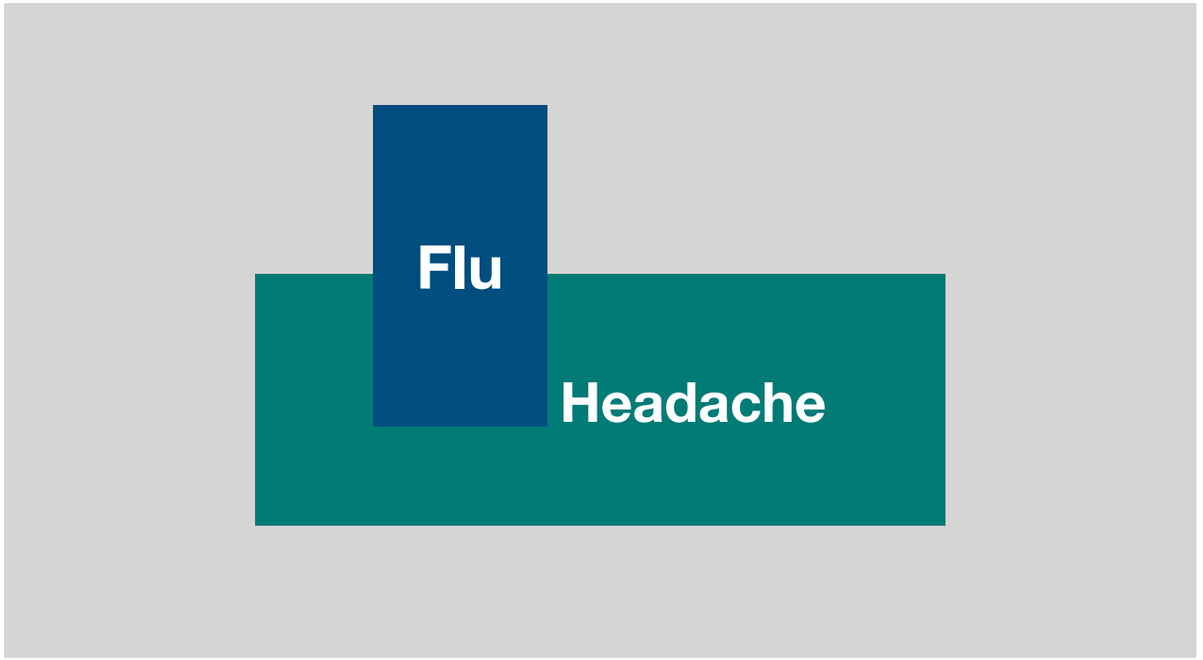


Flu-Headache Example

- *You wake up with a headache. Do you have the flu?*

- ***Priors:***
  - P(H) = 1/10
  - P(F) = 1/40

- P(H | F) = $P(H \wedge F)/P(F)$ = 1/2



Conditional Probability

- **Definition**

$$P(A \mid B) = P(A \wedge B)/P(B)$$

- **Chain rule**

$$P(A \wedge B) = P(A \mid B)P(B)$$

Conditional & Chain

Conditional $P(A \mid B) = P(A \wedge B)/P(B)$

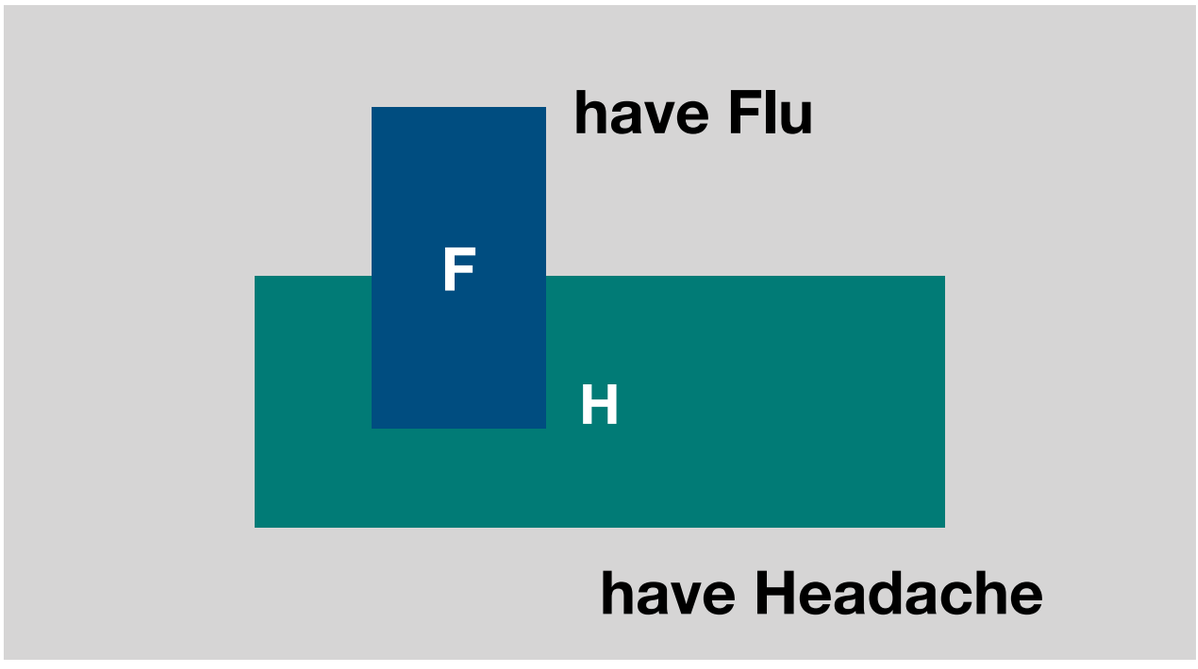
Conditional **Joint** Marginal

Chain $P(A \wedge B) = P(A \mid B)P(B)$

Joint **Conditional** Marginal

Inference

— You wake up with a headache
— You think..“50% of Flu is associated with headaches”
— So ..“50% chance I have Flu”
— Is your reasoning correct?



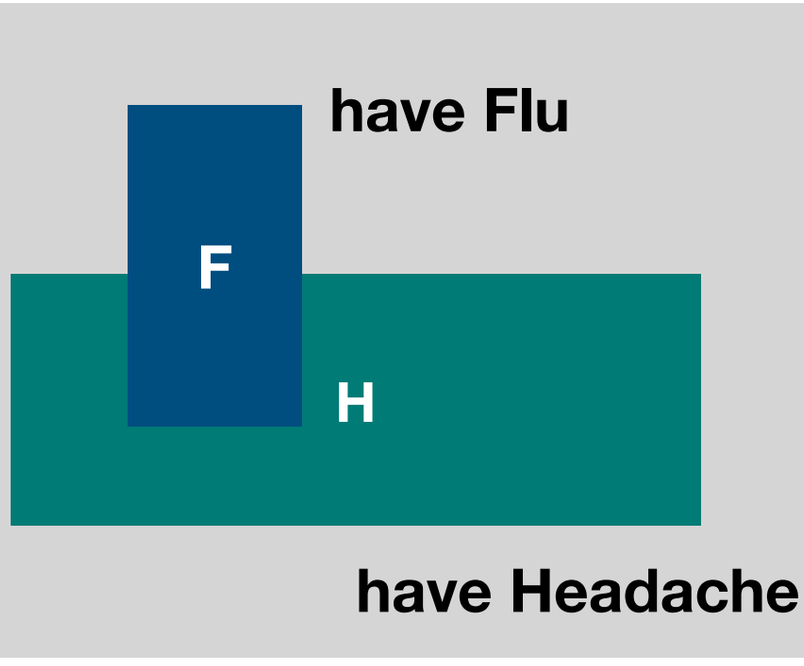
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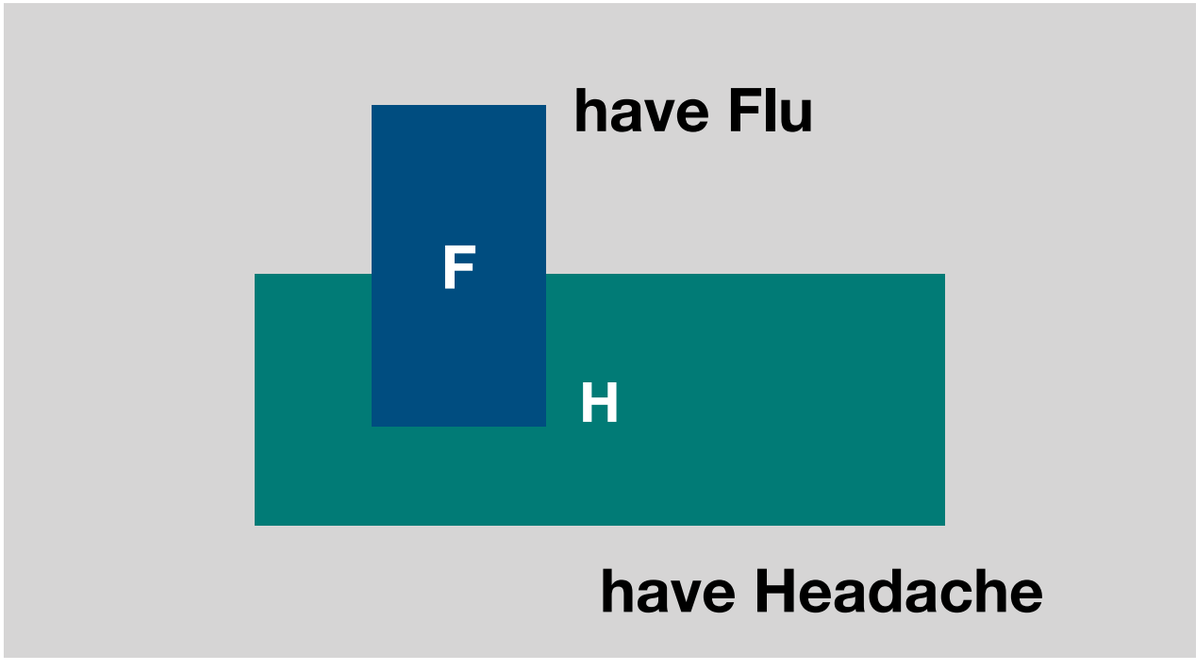
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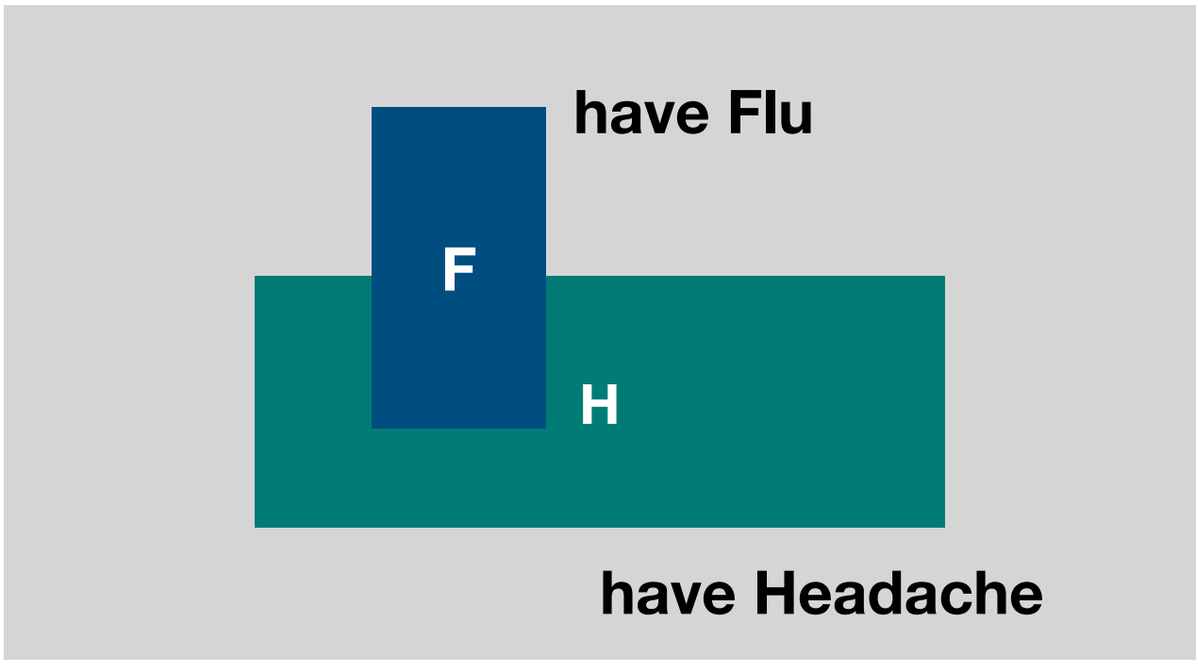
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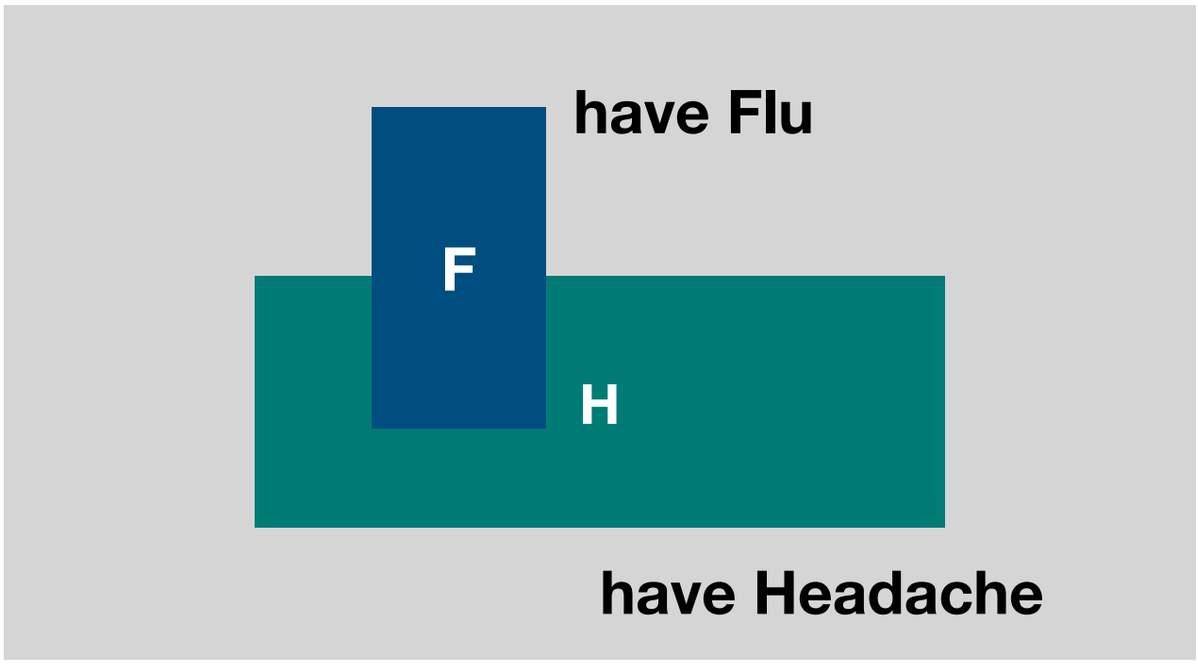
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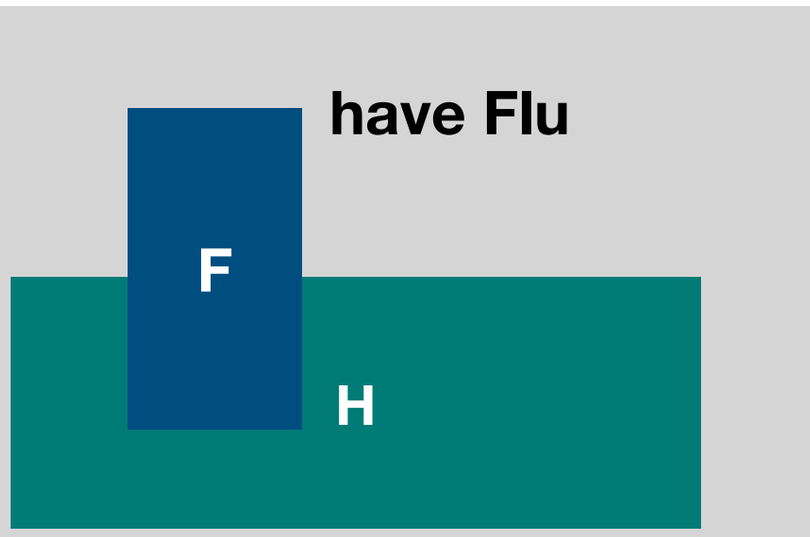
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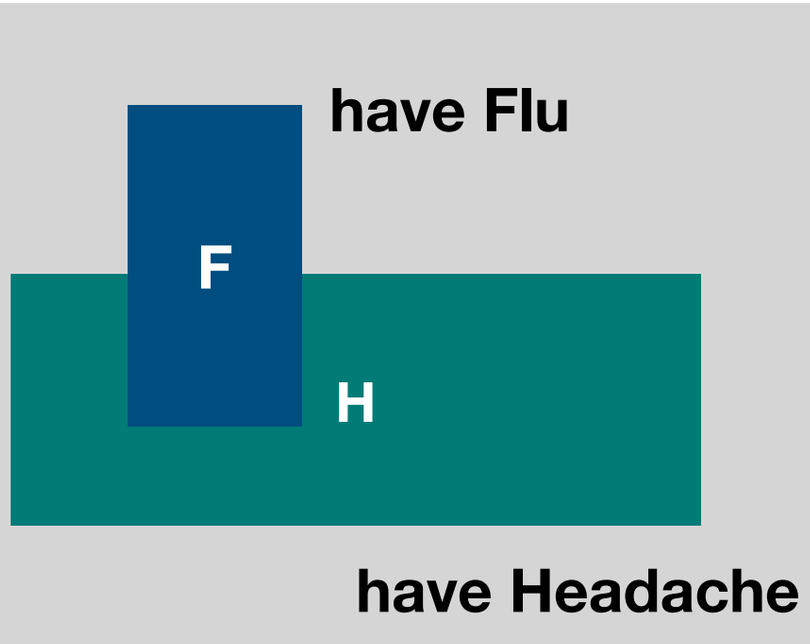
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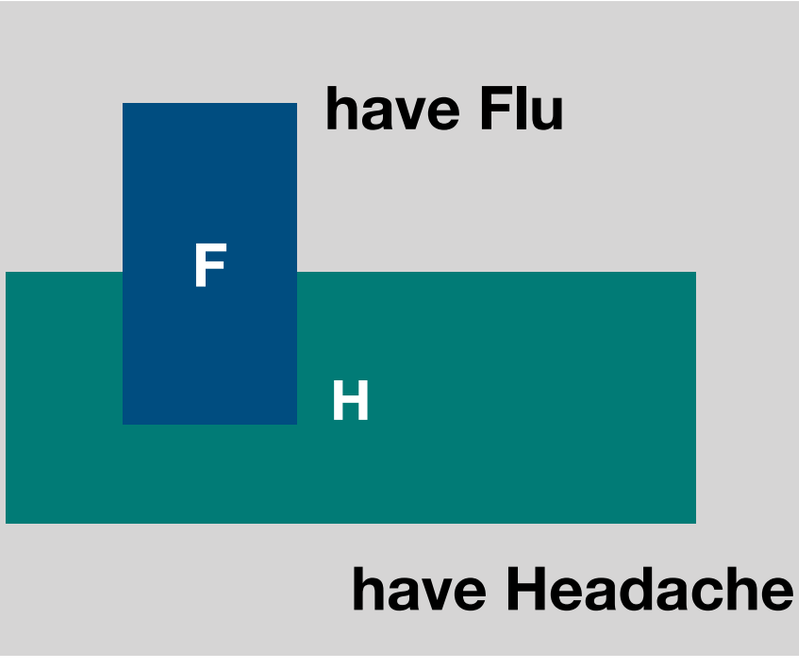
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$$P(F \wedge H) = P(H|F)P(F) = 1/2 * 1/40 = 1/80$$

$$P(F|H) = P(F \wedge H)/P(H) = (1/80)/(1/10) = 1/8$$



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Conditional Distribution Example

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$$P(headache \wedge cold \mid sunny) = P(headache \wedge cold \wedge sunny)/p(sunny)$$

Conditional probability = 0.108 /(0.108 + 0.012 + 0.016 + 0.064)

= 0.54

$$P(headache \wedge cold \mid \sim sunny) = P(headache \wedge cold \wedge \sim sunny)/p(\sim sunny)$$

= 0.072 / (0.072 + 0.008 + 0.144 + 0.576)

= 0.09

Summary

- Probability distributions quantify uncertainty about the world
- Learning reduces uncertainty (by updating distributions)
- **Probability distributions:** joint, marginal, conditional
- **Conditional probability** $P(A \mid B) = P(A \wedge B)/P(B)$
- **Chain rule** $P(A \wedge B) = P(A \mid B)P(B)$

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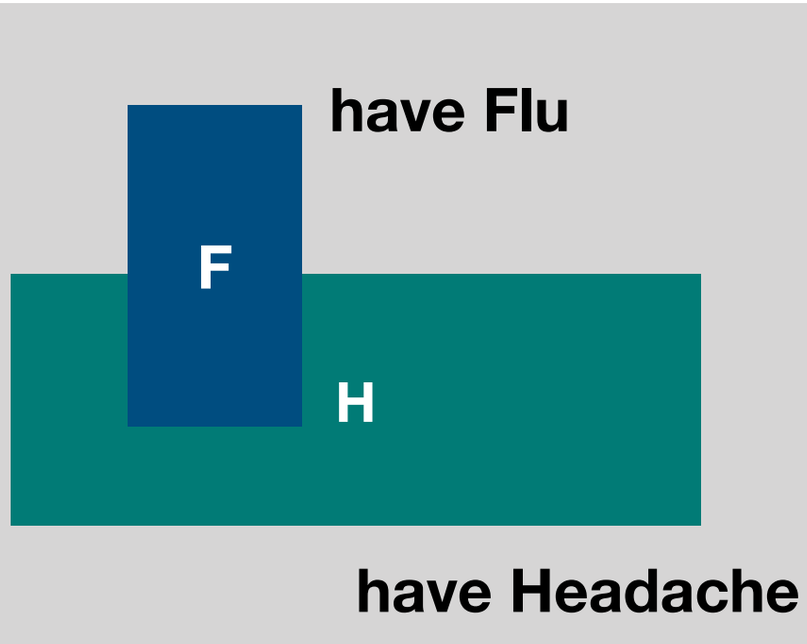
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***P*(H) = 1/10**
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Bayes' Rule

- **Derivation:** from joint probability symmetry

$$P(A \mid B)P(B) = P(A \wedge B) = P(B \wedge A) = P(B \mid A)P(A)$$

- **Bayes' rule**

$$P(B \mid A) = [P(A \mid B)P(B)] \; / \; P(A)$$

- **Interpretation**

  - Prior → Likelihood → Posterior

Conditional, Chain & Bayes

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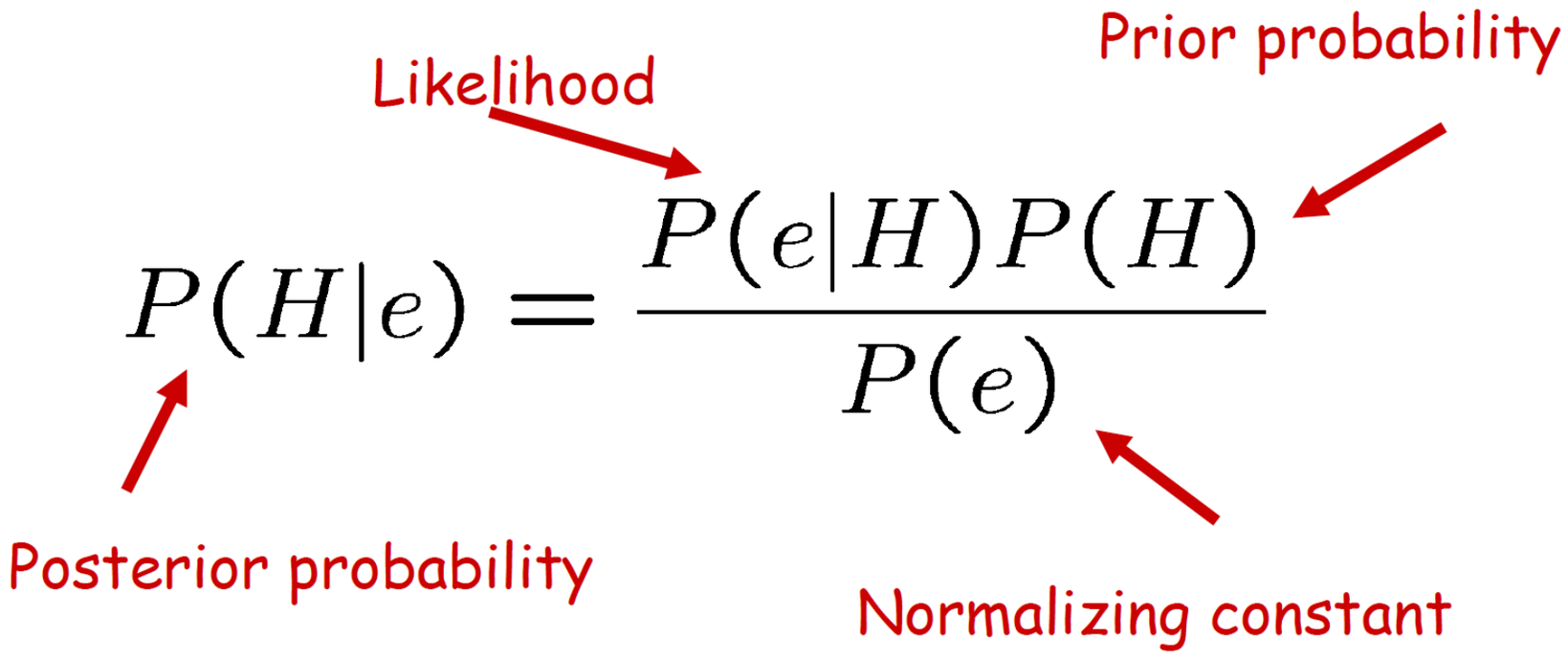
Joint **Conditional** Marginal

Bayes $P(B \mid A) = [P(A \mid B)P(B)] \ / \ P(A)$

Conditional **Conditional** Marginals

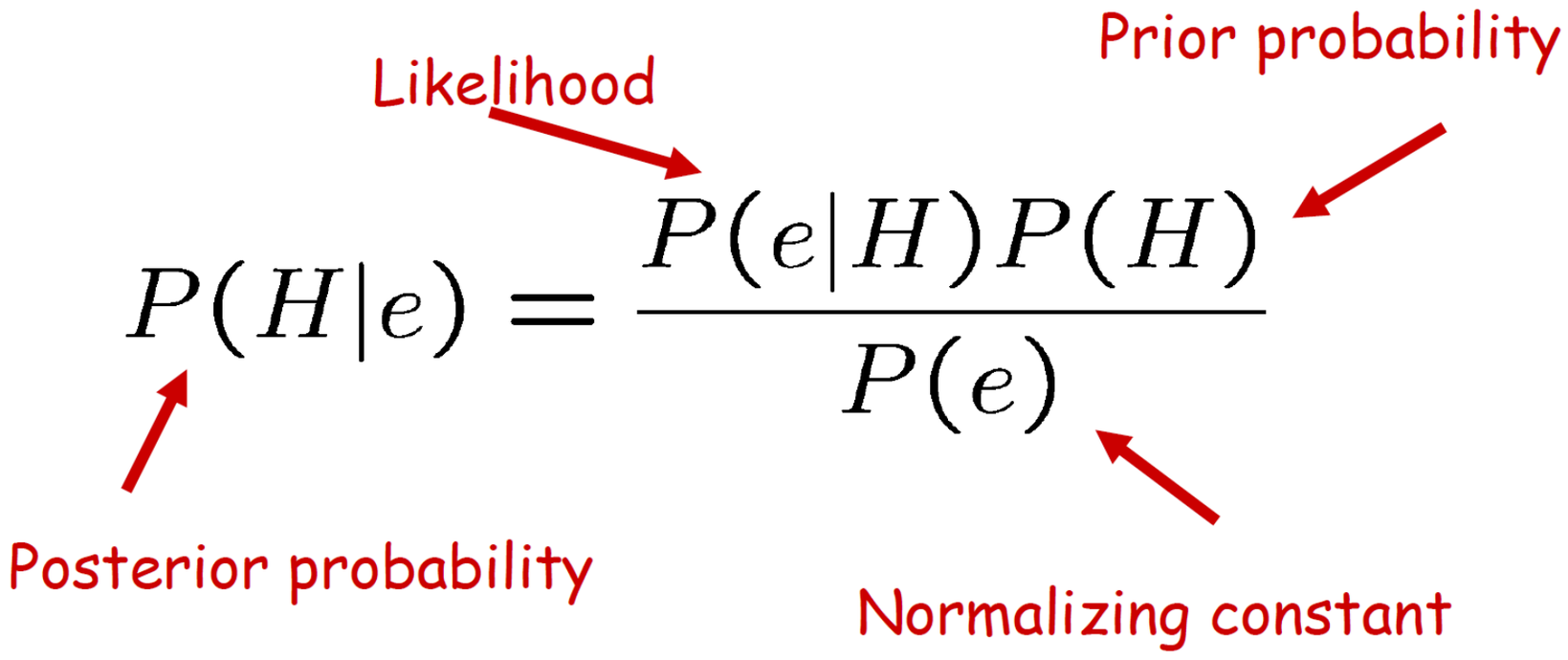
Bayesian Learning

- **Framing ML as Bayesian Learning**
  - Hypotheses = models
  - Data = evidence
  - Learning = posterior update
- **Bayes' rule: states belief of hypothesis H, given evidence e**
- **Posterior: after observing, reduce uncertainty, revised distribution**



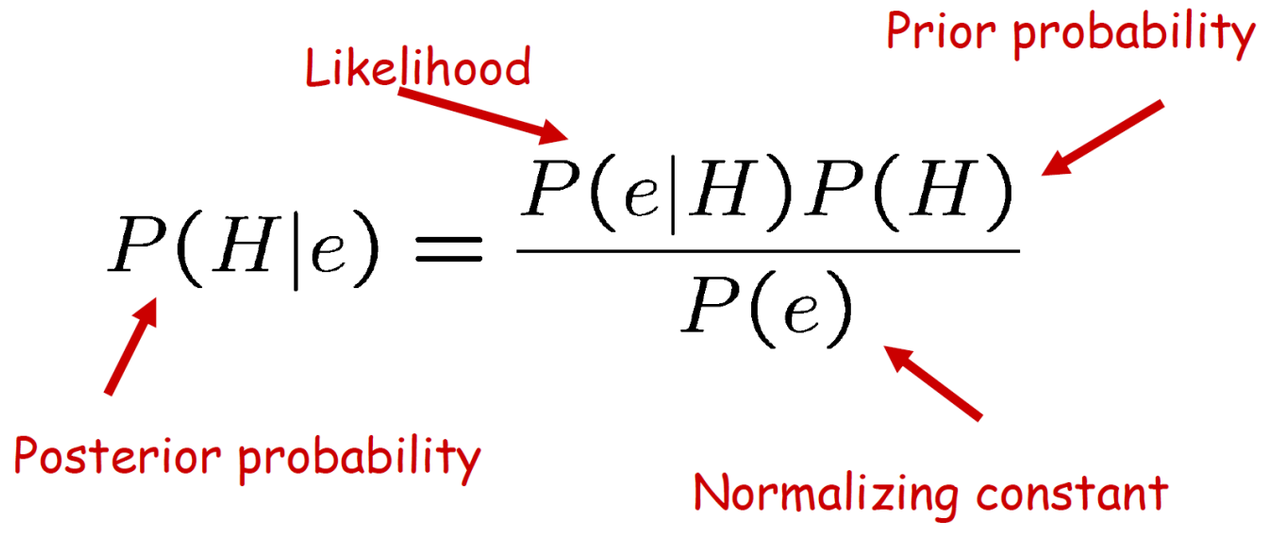
Using Bayes' Rule for Inference

- **Form a hypothesis about the world based on what we observe**
  - Prior of hypotheses: captures prior belief/uncertainty (before observing)
  - Observe data e
  - Likelihood: probability of observing e given H
- **Bayes' rule: states belief of hypothesis H, given evidence e**
- **Posterior: after observing, reduce uncertainty, revised distribution**



Bayesian Learning

- **ML from a statistical perspective:**
  Use Bayes rule to reduce uncertainty
  Find best/correct hypothesis



- — Start with a **Prior** P(H)
  — Observe dataset/**evidence** e = <e_1, e_2, …, e_N>
  — **Likelihood distribution** P(e|H) Likelihood of obtaining a certain dataset (or evidence) given each hypothesis
  — Use Bayes rule to compute a posterior (encodes what's learned)

$$P(H \mid \mathbf{e}) = k\ P(\mathbf{e} \mid H)P(H)$$

- Assume equal probability of e for each hypothesis (normalizing constant k)

Bayesian Prediction/Inference

- To make a prediction about unknown quantity X

$$P(X|\mathbf{e}) = \sum_i P(X|\mathbf{e}, h_i)P(h_i|\mathbf{e})$$

$$= \sum_i P(X|h_i)P(h_i|\mathbf{e})$$

predictions

predictions of individual hypothesis

weight: posterior/belief

h_i: model

weighted average of predictions of individual hypotheses

- Posterior of h is used as a weight of h's prediction
- Prediction: weighted average of predictions of individual hypothesis
- Hypotheses serve as "intermediaries" between raw data & prediction

Candy Example

- Candy sold in two flavors: Lime, Cherry

- Same wrapper for both flavors

- Sold in bags with different ratios:
  – 100% cherry
  – 75% cherry + 25% lime
  – 50% cherry + 50% lime
  – 25% cherry + 75% lime
  – 100% lime

- You bought a bag of candy but don’t know its flavor ratio

- We can run an experiment: eat ***k*** candies: then try to estimate:
  – What’s the flavor ratio of the bag?
  – What flavor will the next candy be?

- What is the hypothesis?

- What is the evidence?

Statistical Learning

- **Hypothesis H:** probabilistic theory of the world
  - h_1: 100% cherry
  - h_2: 75% cherry + 25% lime
  - h_3: 50% cherry + 50% lime
  - h_4: 25% cherry + 75% lime
  - h_5: 100% lime

- **Examples E:** evidence about the world
  - e_1: 1st candy is cherry
  - e_2: 2nd candy is lime
  - e_3: 3rd candy is lime
  - …

Statistical Learning

- Assume prior $P(\mathrm{H})$ = <0.1,0.2,0.4,0.2,0.1>

- Assume candies are **i.i.d. (identically & independently distributed)**

Likelihood distribution: probability of observing a flavor e given a hypothesis h

$$P(\mathbf{e} \mid h) = \prod_n P(e_n \mid h)$$

- Suppose first 10 candies are all lime:

$$P(\mathbf{e} \mid h_5) = 1^{10} = 1$$

$$P(\mathbf{e} \mid h_3) = (1/2)^{10} = 0.00097$$

$$P(\mathbf{e} \mid h_1) = (0)^{10} = 0$$

– h_1: 100% cherry
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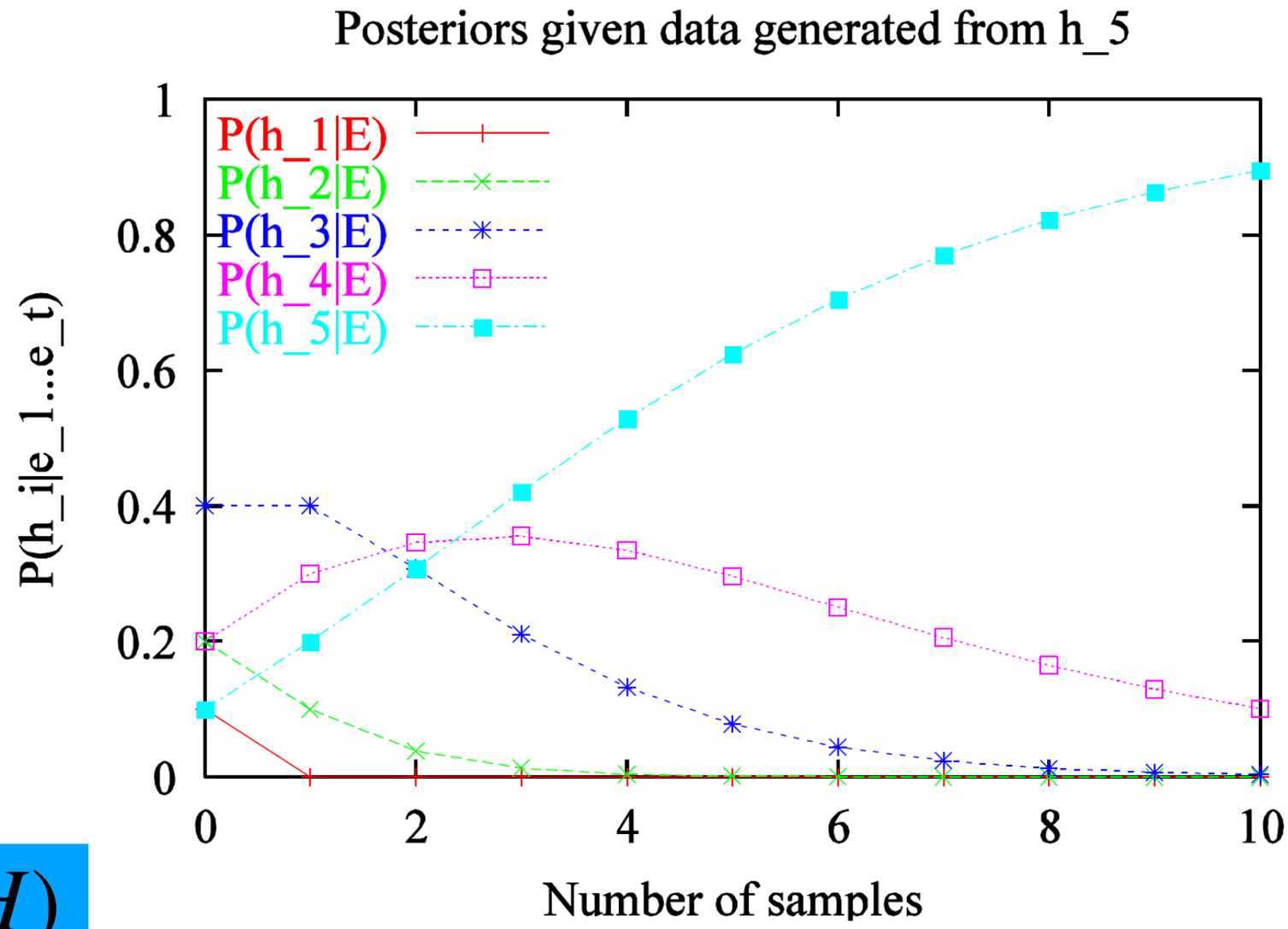
Learning: Compute Posterior

- Hypotheses H:
  - $h1$: 100% cherry
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  - $h4$: 25% cherry + 75% lime
  - $h5$: 100% lime

- Assume prior P(H) = <0.1,0.2,0.4,0.2,0.1>

- True model (that generated training data):
  - h5

- Learning:
  - Update prior beliefs
  - Update distribution given evidence
  - Compute posterior distribution



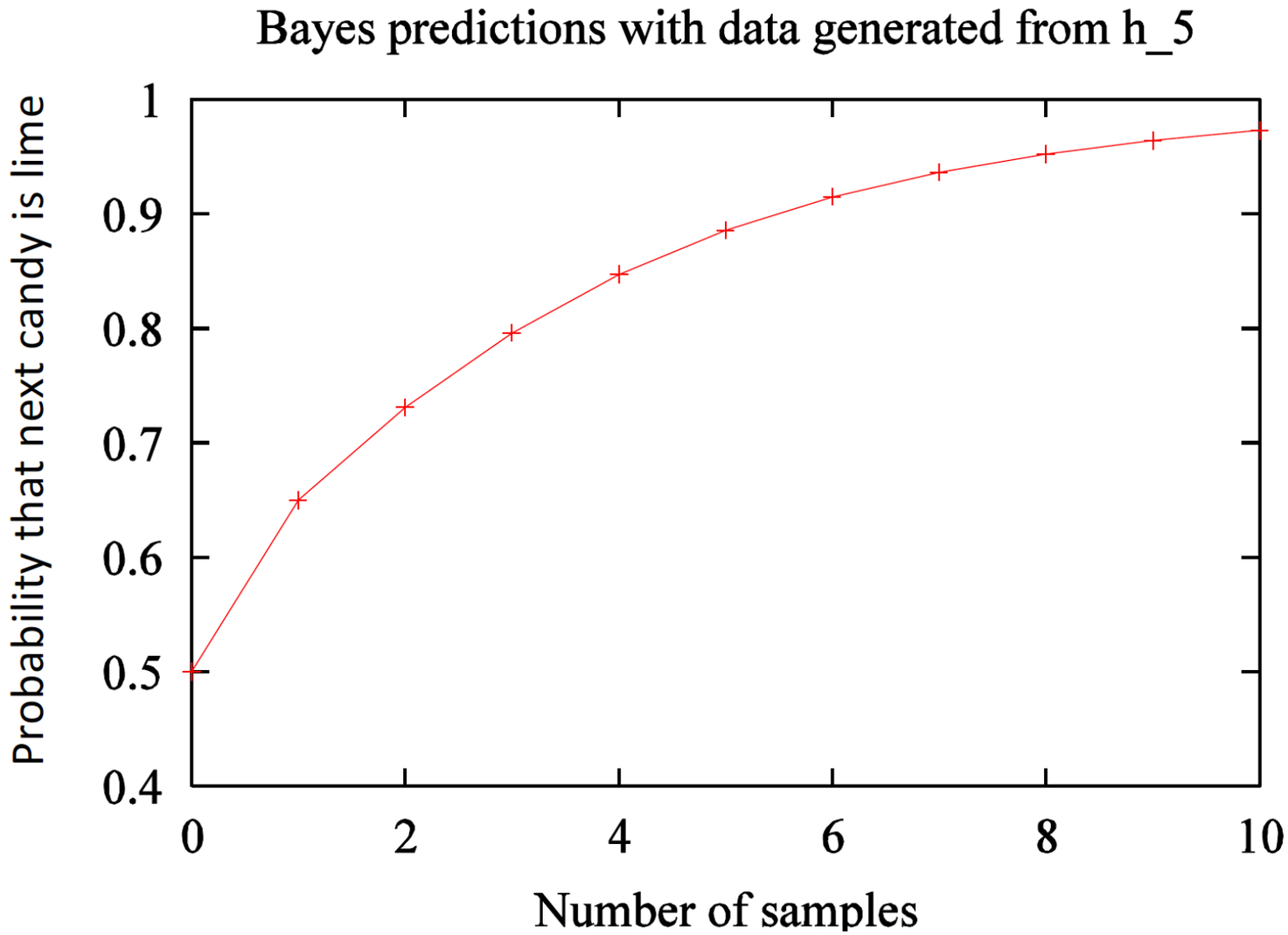
$$P(H \mid \mathbf{e}) = k\ P(\mathbf{e} \mid H)P(H)$$

Inference: predict using posterior

Likelihood probability

$$P(X|\mathbf{e}) = \sum_i P(X|h_i)P(h_i|\mathbf{e})$$

Posterior distribution provides weights for each hypothesis



Bayesian Prediction/Inference

- To make a prediction about unknown quantity X

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Bayesian Learning

- **Properties**
  ++ Optimal (i.e. given prior, no other prediction is correct more often than the Bayesian one)
  ++ **No overfitting** (all hypotheses considered and weighted)

- **Limitation**
  – When hypothesis space is large, may be intractable
  – Sum (or integral) over hypotheses often intractable

- **Solution**
  – “approximate” Bayesian learning (future topic)